



Full iCloud encryption

Apple has announced three new security features to ensure better privacy and data protection. <https://dgit.in/dec22-03>



Anti-Misinformation Project

Google's latest project aims to prevent the spread of misinformation in India. <https://dgit.in/dec22-04>

through different mediums over varying distances while also working on a device that could amplify weak sounds - the Microphone. Though the device that Wheatstone called a microphone was more of a functioning stethoscope, the term would soon be applied to other devices that would be developed for the same purpose later on.

THE FIRST REAL MICROPHONE(SORT OF)

With the Wheatstone microphone already existing, in 1861, German



The Reis Telephone

inventor Johann Philipp Reis, invented a working device that converted sound signals into electric signals which could be propagated through a conductive wire. On the other end, would be a device that would convert the electric signals back into sound making it arguably the first-ever true use of the microphone in the way we define them today: as a transducer that converts mechanical wave energy (sound waves) into electrical energy (audio signals) even though the device was actually an early model of the Telephone, the microphone played a big part in the working function of it.

The Reis' Telephone largely functioned on the movement of a parchment strip through audio waves and then creating disturbances in two brass strips that were connected to the parchment. The moving parchment would result in varying resistances in the circuitry creating the audio signals which would then

be decoded on the receiving end. The Reis telephone is not mentioned a lot these days but is considered by many to be the first ever proper use of a microphone in a device. You may ask why, well to answer that let us take you to the world of Alexander Graham Bell and the Liquid Transmitter device.

BELL AND GRAY'S LIQUID TRANSMITTER

As Bell's Telephone came to be, his idea of a microphone was largely based on Elisha Gray's idea of a



The Liquid Transmitter, which was a major part of Bell's model of the telephone.

microphone using electrolytic fluids to conduct the electric signals of the microphone. Gray had even challenged Bell's patent in court but the court found in Bell's favour much to

Gray's dismay. All of this transpired between the year 1867 and 1876.

In this version, we had a metal cup filled with a mixture of water and sulphuric acid.

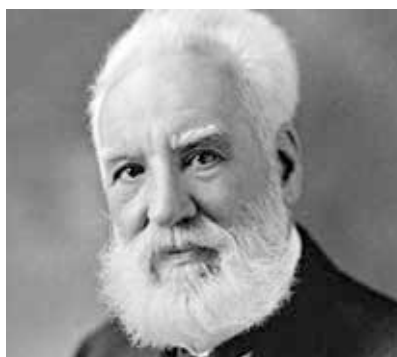
Then, a diaphragm was stretched into a small horn while attached to a needle submerged in the liquid but not touching the metal cup. As the needle would move up and down due to the sound waves hitting the diaphragm, the resistance of the conductive liquid would change.

The changing resistance would be then picked up by another wire submerged in the water which carried an inversely proportional electric signal to the speaker part of the telephone.

This model of the microphone and its success was also very influential in the development of the Carbon Microphone invented by Thomas Edison which we are going to talk about next.

BERLINER AND EDISON'S CARBON MICROPHONE

Let's get to the year 1876 when the first ever proper microphone was created in the form of Berliner and Edison's Carbon Microphone. Edison was much influenced by the ideas and success of the Liquid Transmitter by Bell, and was set on immortalising his own name in the audio transmission industry and immortalise his name, he did. Edison worked alongside Emile Berliner who was a german immigrant in the United States and is said to have done most of the work in the development of the Carbon Microphone.



Alexander Graham Bell



The Carbon Microphone